IGCSE 0607 Practice Questions — E5.5 — Combined Questions & Markscheme

Angles

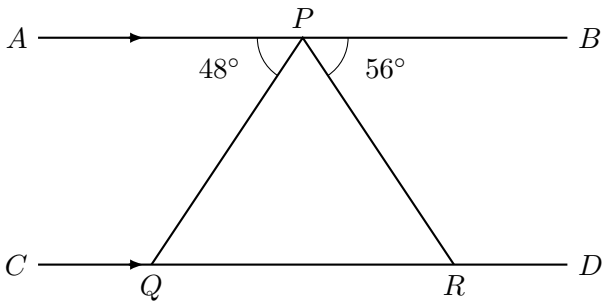
Total Questions: 21 — **Total Marks:** 135 Marks

Table of Contents & Quick Navigation

- **E5.5** Angles
  → **Paper 2 (Non-Calculator)** *(11 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

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1 In the diagram, the line AB is parallel to the line CD.
P is a point on AB.
Q and R are points on CD.
The line PQ and the line PR are drawn.
Angle $APQ = 48°$ and angle $BPR = 56°$.



NOT TO SCALE

(a) Write down the value of angle PQR.

Angle $PQR =$ ..... ° **[1]**

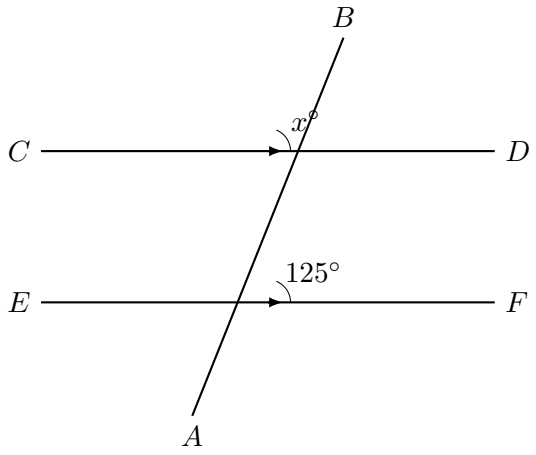
(b) Write down the value of angle PRQ.

Angle $PRQ =$ ..... ° **[1]**

(c) Work out the value of angle QPR.

Angle $QPR =$ ..... ° **[1]**

2 The diagram shows a straight line AB intersecting two parallel lines, CD and EF.



(a) Find the value of x.

$x =$ ..... **[1]**

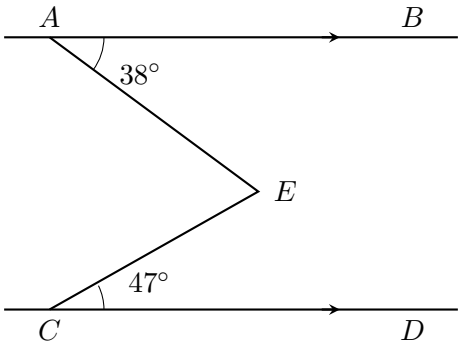
(b) Give a geometric reason for your answer to part (a).

..... **[1]**

(c) Calculate the interior angle of a regular decagon (10-sided polygon).

..... ° **[1]**

3 In the diagram, the line AB is parallel to the line CD.
The point E lies between the two parallel lines.



NOT TO SCALE

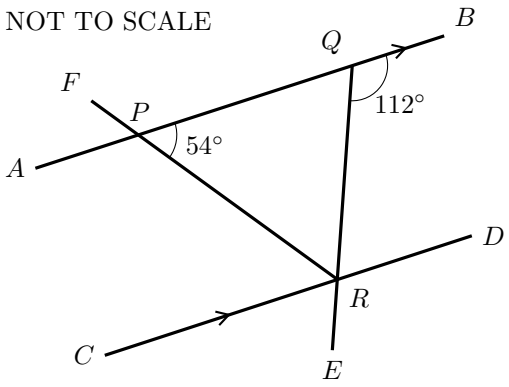
(a) Calculate the size of angle AEC.

Angle AEC = ... ° **[2]**

(b) A regular polygon has an exterior angle of 24°.
Work out the number of sides of this polygon.

..... **[2]**

4 In the diagram, AB is parallel to CD.
The straight line FPR intersects AB at P and CD at R.
The straight line QRE intersects AB at Q and CD at R.
Angle $QPR = 54°$ and angle $BQR = 112°$.



(a) Write down the size of angle PRC and give a geometrical reason for your answer.

Angle $PRC =$ ..... , Reason: ..... **[2]**

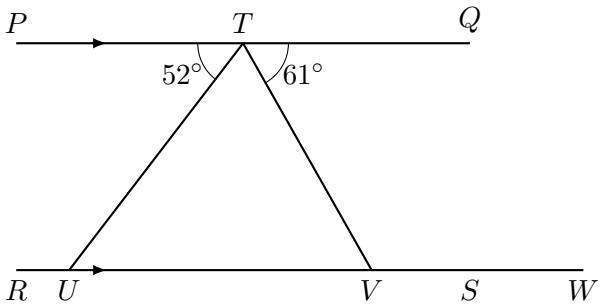
(b) Calculate the size of angle PRQ.

Angle $PRQ =$ ..... ° **[2]**

(c) Write down the size of angle CRE.

Angle $CRE =$ ..... ° **[1]**

5 In the diagram, the line PQ is parallel to the line RS.
The point T lies on PQ, and the points U and V lie on RS.
The line RS is extended to a point W.
Angle $PTU = 52°$ and angle $QTV = 61°$.



(a) Calculate angle TUV.

Angle TUV = ... ° **[1]**

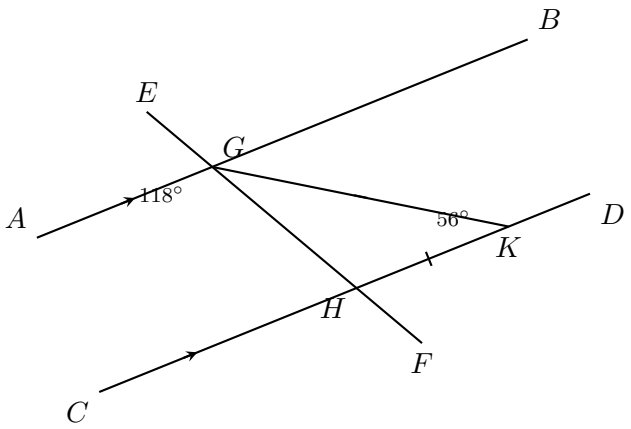
(b) Calculate angle UTV.

Angle UTV = ... ° **[2]**

(c) Calculate angle TVW.

Angle TVW = ... ° **[2]**

6 In the diagram, AB is parallel to CD.
The straight line EF intersects AB at G and CD at H.
Point K lies on CD such that $GK = HK$, $\angle AGH = 118°$ and $\angle GKH = 56°$.



(a) Find the size of angle GHD, giving a geometrical reason for your answer.

$\angle GHD =$ ..... ° **[2]**

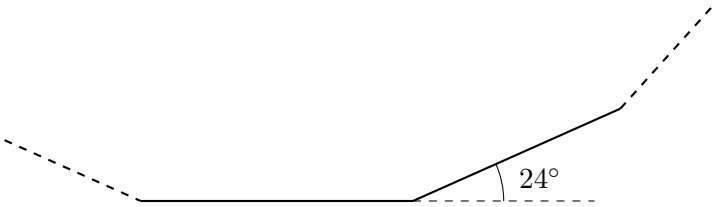
(b) Calculate the size of angle GHK.

$\angle GHK =$ ..... ° **[1]**

(c) Calculate the size of angle BGK.

$\angle BGK =$ ..... ° **[2]**

7 The diagram shows part of a regular polygon with n sides.
The exterior angle of this polygon is 24°.



(a) Calculate the value of n.

$n =$ ..... **[2]**

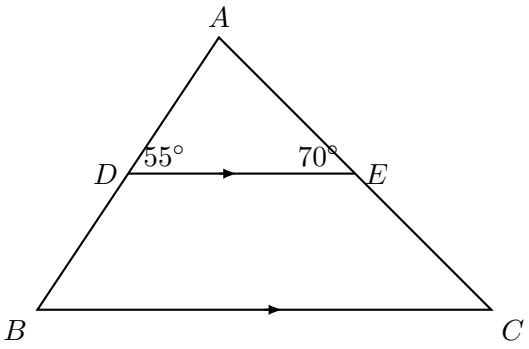
(b) Calculate the sum of the interior angles of this polygon.

..... ° **[2]**

(c) A different regular polygon has an interior angle of 165°.
Work out the number of sides of this polygon.

..... **[2]**

8 The diagram shows a triangle ABC with a line DE drawn parallel to BC.
D lies on AB and E lies on AC.
Angle $ADE = 55°$ and angle $AED = 70°$.



(a) Calculate angle DAE.

Angle $DAE =$ ..... ° **[2]**

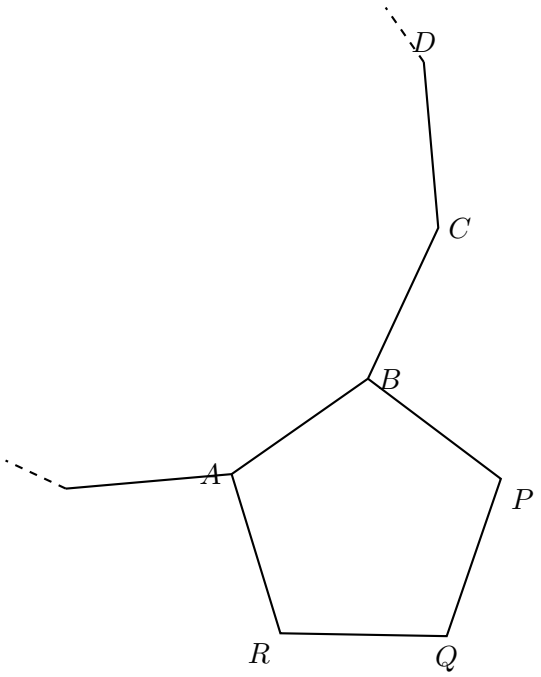
(b) Calculate angle ABC.

Angle $ABC =$ ..... ° **[1]**

(c) Explain why triangle ADE is an isosceles triangle.

[2]

9 The diagram shows part of a regular n-sided polygon $ABCDE\ldots$ and a regular pentagon $ABPQR$ sharing the common side AB.



(a) Calculate the size of one interior angle of the regular pentagon $ABPQR$.

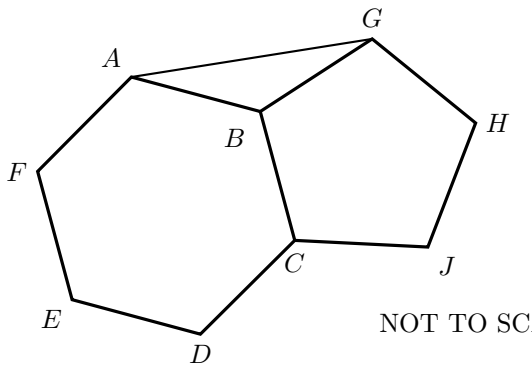
..... ° **[2]**

(b) The reflex angle CBP is 102°.

Calculate the value of n, the number of sides of the regular polygon $ABCDE\ldots$.

$n =$ ..... **[4]**

10 The diagram shows a regular hexagon $ABCDEF$ and a regular pentagon $BCGHJ$ sharing a common side BC.
Vertices A and G are joined by a straight line.



NOT TO SCALE

(a) Calculate the size of

(i) interior angle ABC of the regular hexagon,

Angle ABC = ... ° **[2]**

(ii) interior angle CBG of the regular pentagon.

Angle CBG = ... ° **[1]**

(b) Calculate the size of angle BAG.

Angle BAG = ... ° **[3]**

(c) An irregular quadrilateral has interior angles $(2x + 10)°$, $(3x - 15)°$, $(x + 25)°$, and $100°$.
Calculate the size of the largest interior angle of this quadrilateral.

..... ° **[3]**

11 Answer the following questions about polygons and angles.

(a) An irregular polygon has n sides.
Three of the interior angles are each $140°$.
The remaining $(n-3)$ interior angles are each $165°$.

Work out the value of n.

$n =$ ..... **[4]**

(b) In triangle ABC, the angles are $\angle A = a°$, $\angle B = b°$ and $\angle C = c°$.
It is given that $c - b = a$.

Prove that ABC is a right-angled triangle.

[2]

(c) Polygon P is a regular polygon with 12 sides.
Polygon Q is a regular polygon with k sides.
The exterior angle of Polygon Q is half the size of the exterior angle of Polygon P.
Polygon P and Polygon Q share a common edge AB.

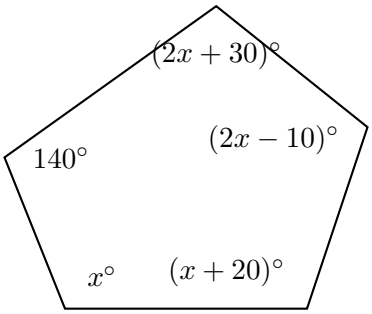
(i) Find the value of k.

$k =$ ..... **[2]**

(ii) Work out the size of the angle at vertex B that lies outside both polygons.

..... ° **[2]**

12 The diagram shows an irregular pentagon.
The interior angles of the pentagon are $x°$, $(x + 20)°$, $(2x - 10)°$, $(2x + 30)°$ and $140°$.



NOT TO SCALE

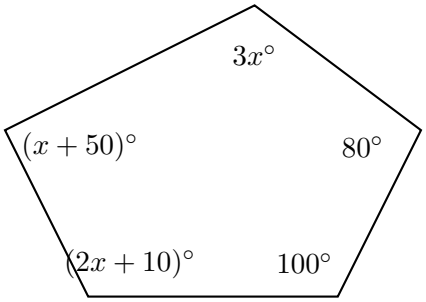
(a) Calculate the value of x.

$x =$ ..... **[3]**

(b) Find the size of the largest interior angle of the pentagon.

..... ° **[2]**

13 The diagram shows an irregular pentagon.



NOT TO SCALE

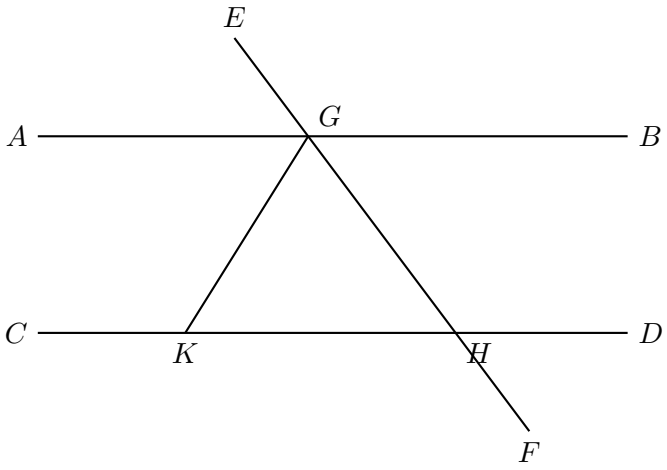
(a) Work out the value of x.

$x =$ ..... **[3]**

(b) Another regular polygon has an interior angle that is 4 times the size of its exterior angle.
Calculate the number of sides of this polygon.

..... **[3]**

14 In the diagram, the line AB is parallel to the line CD.
The transversal EF intersects AB at G and CD at H.
The point K lies on CD.



(a) Angle $EGB = (5x + 10)°$ and angle $GHD = (7x - 30)°$.
Find the value of x.
Give a geometrical reason for your working.

$x =$ ..... , Reason: ..... **[3]**

(b) Angle $HGK = 35°$.
Calculate angle GKH.

..... ° **[3]**

15 **(a)** This question is about the interior and exterior angles of polygons.

(i) The interior angles of a hexagon are $x°$, $2x°$, $(x + 30)°$, $(2x + 10)°$, $(3x - 20)°$, and $160°$.
Work out the value of x.

$x =$ ..... **[3]**

(ii) A regular polygon has an interior angle that is 6.5 times its exterior angle.
Calculate the number of sides of this polygon.

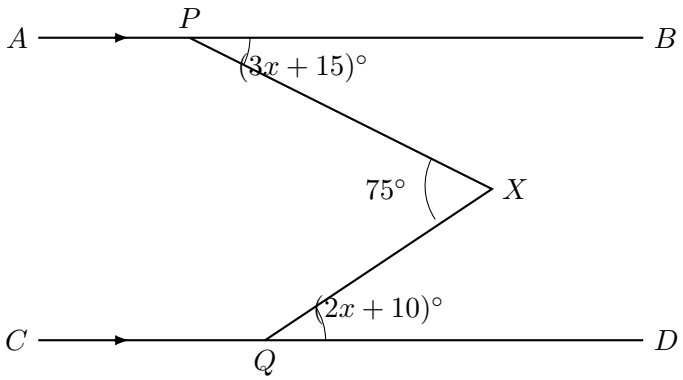
..... **[3]**

(iii) Two regular polygons, P and Q, have m sides and n sides respectively.
The sum of their exterior angles is $54°$.
The difference between their interior angles is $18°$.
Find the value of m and the value of n.

$m =$ ..... , $n =$ ..... **[5]**

16 **(a)** In the diagram, the line AB is parallel to the line CD.
X is a point between AB and CD.
The point P is on AB and the point Q is on CD.
Angle $BPX = (3x + 15)°$, angle $DQX = (2x + 10)°$ and angle $PXQ = 75°$.

Find the value of x.



NOT TO SCALE

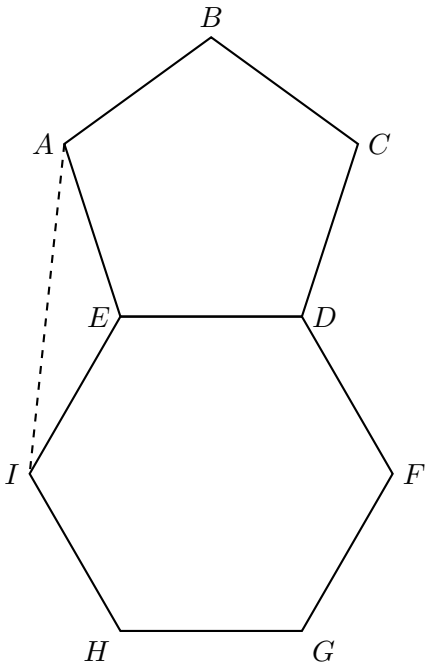
$x =$ ..... **[3]**

(b) Polygon P is a regular polygon with n sides.
Polygon Q is a regular polygon with $3n$ sides.
The size of an interior angle of polygon Q is 16° greater than the size of an interior angle of polygon P.

Calculate the value of n.

$n =$ ..... **[4]**

17 The diagram shows a regular pentagon $ABCDE$ and a regular hexagon $EDFGHI$ sharing a common edge ED.



(a) Calculate angle CDE.

Angle CDE = ... ° **[2]**

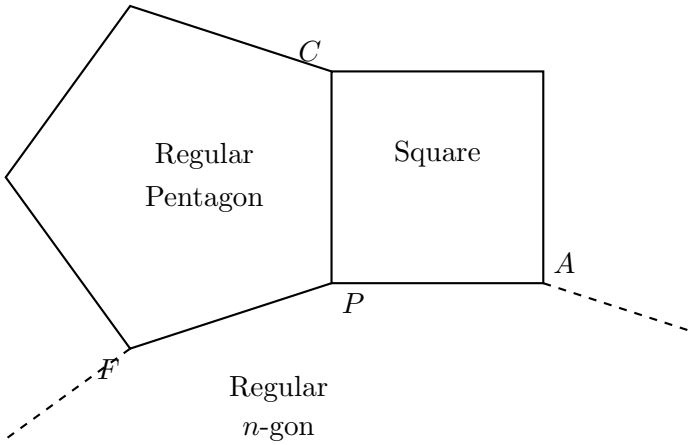
(b) Calculate angle AEI.

Angle AEI = ... ° **[2]**

(c) Calculate angle EAI.

Angle EAI = ... ° **[3]**

18 The diagram shows three regular polygons meeting at the point P.
The polygons are a square, a regular pentagon, and a regular polygon with n sides.
The square shares the side PC with the pentagon and the side PA with the n-gon.
The pentagon shares the side PF with the n-gon.
They meet with no gaps and do not overlap.



(a) Calculate the interior angle of a regular pentagon.

..... ° **[2]**

(b) Find the value of n.

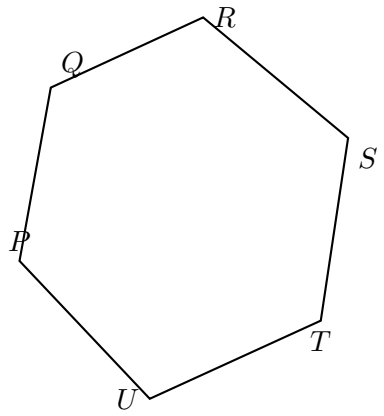
$n =$ ..... **[3]**

(c) The vertices A and F of the n-gon are joined by a straight line.
Calculate angle PFA.

..... ° **[2]**

19 The diagram shows an irregular hexagon $PQRSTU$.
The interior angles of the hexagon are:

- $\angle P = (2x + 15)°$
- $\angle Q = (3x - 20)°$
- $\angle R = (2x + 35)°$
- $\angle S = (4x - 45)°$
- $\angle T = (x + 50)°$
- $\angle U = (3x + 10)°$



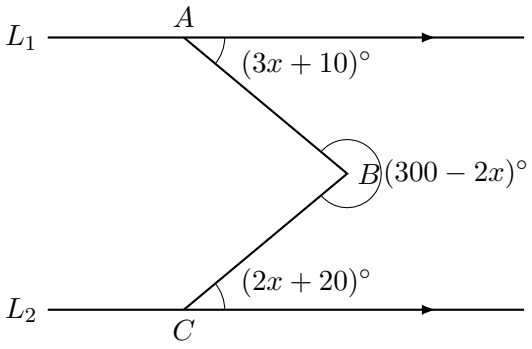
(a) Calculate the value of x.

$x =$ ..... **[4]**

(b) Determine whether PQ is parallel to TS, showing clear geometrical working to support your answer.

..... **[3]**

20 The diagram shows two parallel lines, L_1 and L_2.
The point A lies on L_1 and the point C lies on L_2.
The point B lies between the two parallel lines.
The acute angle between L_1 and AB is $(3x + 10)°$.
The acute angle between L_2 and CB is $(2x + 20)°$.
The reflex angle ABC is $(300 - 2x)°$.



(a) Show that $x = 10$.

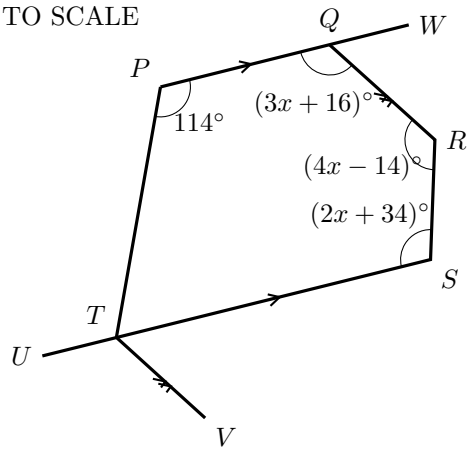
[4]

(b) The points A, B, and C form a triangle ABC.
Given that $AB = BC$, calculate the angle BCA.

Angle BCA = ... ° **[3]**

21 The diagram shows an irregular pentagon $PQRST$.
The side PQ is extended to W and the side ST is extended to U.
PW is parallel to US, and QR is parallel to TV.
Angle $QPT = 114°$, angle $PQR = (3x + 16)°$, angle $QRS = (4x - 14)°$, and angle $RST = (2x + 34)°$.

NOT TO SCALE



(a) Find the size of angle PTS and give a geometrical reason for your answer.

Angle $PTS =$ ..... , Reason: ..... **[2]**

(b) Show that $x = 36$.

[3]

(c) Calculate the size of

(i) angle PTU,

Angle $PTU =$ ..... ° **[1]**

(ii) angle STV.

Angle $STV =$ ..... ° **[2]**

(d) A regular polygon has n sides.
Each interior angle of this regular polygon is 11 times the size of each exterior angle.
Calculate the value of n.

$n =$ ..... **[3]**

1 **(a)** 48 **[B1]**

(b) 56 **[B1]**

(c) $180 - 48 - 56 = 76$ **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 125 **[B1]**

(b) Corresponding angles (are equal) **[B1]**

(c) 144 **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** Identifies alternate angles by drawing a parallel line through E OR forms a triangle by extending a line **[M1]**

85 **[A1]**

(b) $\frac{360}{24}$ **[M1]**

15 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Angle $PRC = 54°$ **[B1]**

Reason: alternate angles (AB is parallel to CD) **[B1]**

(b) Angle $PQR = 180° - 112° = 68°$ **[M1]**

Angle $PRQ = 180° - 54° - 68° = 58$ **[A1]**

(c) 58 **[B1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** 52 **[B1]**

(b) Angles on a straight line sum to 180°: $180 - (52 + 61)$ **[M1]**

67 **[A1]**

(c) Identifies angle $TVU = 61°$ (alternate angle to QTV) or angle $VTW + TVW$ exterior relationship **[M1]**

Angle $TVW = 180 - 61 = 119$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** 118 **[B1]**

Alternate angles (are equal) **[B1]**

(b) $180° - 118° = 62$ **[B1]**

(c) Uses isosceles triangle GHK: $\angle HGK = 62°$ and $\angle BGH = 180° - 118° = 62°$ (or $\angle BGK = \angle GKD = 180° - 56°$) **[M1]**

124 **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** $360 \div 24$ **[M1]**

15 **[A1]**

(b) $(15 - 2) \times 180$ **[M1]**

2340 **[A1]**

(c) Exterior angle $= 180 - 165 = 15°$ **[M1]**

$360 \div 15 = 24$ **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** $180 - (55 + 70)$ **[M1]**

55 **[A1]**

(b) 55 **[B1]**

(c) Angle $DAE = 55°$ (from part a) AND Angle $ADE = 55°$ **[B1]**

Two angles are equal, so the triangle is isosceles **[B1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** Exterior angle $= \dfrac{360°}{5} = 72°$ or interior sum $= (5 - 2) \times 180° = 540°$ **[M1]**

108 **[A1]**

(b) Uses angles at point B: $\angle ABC = 360° - 108° - 102°$ **[M1]**

$\angle ABC = 150°$ **[A1]**

Exterior angle of regular n-gon $= 180° - 150° = 30°$ **[M1]**

$n = \dfrac{360°}{30°} = 12$ **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** **(i)** $\dfrac{(6 - 2) \times 180°}{6}$ or $180° - \dfrac{360°}{6}$ **[M1]**

120 **[A1]**

(ii) 108 **[B1]**

(b) Angle $ABG = 360° - 120° - 108° = 132°$ **[M1]**

Triangle ABG is isosceles ($AB = BG$) **[M1]**

$\frac{180° - 132°}{2} = 24$ **[A1]**

(c) $(2x + 10) + (3x - 15) + (x + 25) + 100 = 360$ **[M1]**

$6x + 120 = 360$ giving $x = 40$ **[A1]**

Largest angle $= 3(40) - 15 = 105$ **[A1]**

[Back to Question] *[Total Marks: 9]*

11 **(a)** Equates sum of angles to formula: $3 \times 140 + 165(n - 3) = 180(n - 2)$ **[M1]**

Expands brackets correctly: $420 + 165n - 495 = 180n - 360$ **[M1]**

Rearranges to isolate n: $15n = 285$ (or equivalent) **[M1]**

19 **[A1]**

(b) States $a + b + c = 180$ **[B1]**

Substitutes $a + b = c$ to get $c + c = 180 \implies 2c = 180 \implies c = 90$ completing the proof **[B1]**

(c) **(i)** Calculates exterior angle of $P = \frac{360}{12} = 30°$ so exterior angle of $Q = 15°$ **[M1]**

24 **[A1]**

(ii) Calculates interior angles of P and Q: $180 - 30 = 150°$ and $180 - 15 = 165°$ **[M1]**

$360 - (150 + 165) = 45$ **[A1]**

[Back to Question] *[Total Marks: 10]*

12 **(a)** Sum of interior angles of a pentagon $= (5 - 2) \times 180° = 540°$ **[M1]**

$x + (x + 20) + (2x - 10) + (2x + 30) + 140 = 540$ **[M1]**

$6x + 180 = 540 \implies 6x = 360 \implies x = 60$ **[A1]**

(b) Substituting $x = 60$ into $(2x + 30)$ or checking all angle sizes $(60°, 80°, 110°, 150°, 140°)$ **[M1]**

150 **[A1]**

[Back to Question] *[Total Marks: 5]*

13 **(a)** Recognises the sum of the interior angles of a pentagon is $540°$ **[B1]**

Sets up a correct equation: $(2x + 10) + 100 + 80 + 3x + (x + 50) = 540$ **[M1]**

50 **[A1]**

(b) Sets up a relationship for the angles, e.g., $4E + E = 180$ where E is the exterior angle **[M1]**

$E = 36°$ **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** $5x + 10 = 7x - 30$ **[M1]**

$x = 20$ **[A1]**

Corresponding angles (are equal) **[B1]**

(b) Angle $GHD = 7(20) - 30 = 110°$ **[B1]**

Angle $GHK = 180 - 110 = 70°$ (Angles on a straight line) **[B1]**

Angle $GKH = 180 - (70 + 35) = 75°$ (Angles in a triangle) **[B1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** **(i)** Equates the sum of the angles to the interior angle sum of a hexagon (720°):
$x + 2x + x + 30 + 2x + 10 + 3x - 20 + 160 = 720$ **[M1]**

Simplifies the equation to $9x + 180 = 720$ (or $9x = 540$) **[M1]**

60 **[A1]**

(ii) Sets up an equation for the exterior angle e:
$6.5e + e = 180°$ **[M1]**

$7.5e = 180° \implies e = 24°$ **[M1]**

Number of sides $n = \dfrac{360}{24} = 15$ **[A1]**

(iii) Let exterior angles be E_m and E_n. Sets up the first equation:
$E_m + E_n = 54$ **[B1]**

Relates interior difference to exterior difference:
$(180 - E_m) - (180 - E_n) = 18 \implies E_n - E_m = 18$ **[M1]**

Solves the simultaneous equations by addition (or substitution):
$2E_n = 72 \implies E_n = 36°$ **[M1]**

Calculates n: $n = \dfrac{360}{36} = 10$ **[A1]**

Calculates m: $E_m = 18° \implies m = \dfrac{360}{18} = 20$ **[A1]**

[Back to Question] *[Total Marks: 11]*

16 **(a)** Recognises that $\angle PXQ = \angle BPX + \angle DQX$ (e.g. by drawing a parallel line through X to form alternate angles) **[M1]**

$(3x + 15) + (2x + 10) = 75$ **[M1]**

$5x + 25 = 75 \implies 5x = 50 \implies x = 10$ **[A1]**

(b) Expression for exterior or interior angle of P: $\dfrac{360}{n}$ or $180 - \dfrac{360}{n}$ **[B1]**

Expression for exterior or interior angle of Q: $\dfrac{360}{3n}$ or $180 - \dfrac{120}{n}$ **[B1]**

Forms a correct equation: $\left(180 - \dfrac{120}{n}\right) - \left(180 - \dfrac{360}{n}\right) = 16$ OR $\dfrac{360}{n} - \dfrac{120}{n} = 16$ **[M1]**

$\dfrac{240}{n} = 16 \implies n = 15$ **[A1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** $\frac{(5-2)\times 180}{5}$ OR $180-\frac{360}{5}$ **[M1]**

108 **[A1]**

(b) Interior angle of hexagon $= 120°$ **[B1]**

Angle $AEI = 360 - 108 - 120 = 132$ **[A1]**

(c) $AE = EI$ (sides of regular polygons sharing an edge are equal) **[M1]**

Triangle AEI is isosceles **[M1]**

Angle $EAI = \frac{180-132}{2} = 24$ **[A1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** $(5-2)\times 180 \div 5$ OR $180-(360\div 5)$ **[M1]**

108 **[A1]**

(b) Interior angle of square $= 90°$ **[B1]**

Interior angle of n-gon $= 360 - 90 - 108 = 162°$ **[M1]**

$n = 360 \div (180 - 162) = 20$ **[A1]**

(c) Recognises triangle PFA is isosceles (since $PA = PF$ from shared sides of regular polygons) **[M1]**

Angle $PFA = (180 - 162) \div 2 = 9$ **[A1]**

[Back to Question] *[Total Marks: 7]*

19 **(a)** Calculates interior angle sum of a hexagon: $(6-2)\times 180° = 720°$ **[B1]**

Sums algebraic expressions:
$(2x+15)+(3x-20)+(2x+35)+(4x-45)+(x+50)+(3x+10) = 720$ **[M1]**

Simplifies to $15x + 45 = 720$ **[M1]**

45 **[A1]**

(b) Calculates $\angle Q = 3(45) - 20 = 115°$, $\angle R = 2(45) + 35 = 125°$, and $\angle S = 4(45) - 45 = 135°$ **[M1]**

Finds sum of interior angles along QRS: $115° + 125° + 135° = 375°$ (or constructs line through R parallel to PQ, giving co-interior angles $65° + 60° = 125° e 135°$) **[M1]**

States No, because the sum of interior angles $\angle Q + \angle R + \angle S = 375° e 360°$ (or corresponding/alternate angles are not equal) **[A1]**

[Back to Question] *[Total Marks: 7]*

20 **(a)** Finds an expression for the interior angle ABC using the sum of angles at a point:
Interior angle $ABC = 360° - (300 - 2x)° = (60 + 2x)°$ **[M1]**

Draws a parallel line through B and uses alternate angles to form an equation:
Interior angle $ABC = (3x + 10)° + (2x + 20)°$ **[M1]**

Equates the two expressions for the interior angle ABC:

$5x + 30 = 60 + 2x$ **[M1]**

Solves the equation correctly to show the result:
$3x = 30 \implies x = 10$ **[A1]**

(b) Calculates the interior angle ABC using $x = 10$:
Angle $ABC = 60 + 2(10) = 80°$ **[M1]**

Recognises triangle ABC is isosceles and sets up the base angle calculation:
$\frac{180 - 80}{2}$ **[M1]**

50 **[A1]**

[Back to Question] *[Total Marks: 7]*

21 **(a)** Angle $PTS = 66°$ **[B1]**

Reason: co-interior angles sum to 180° (PW is parallel to US) **[B1]**

(b) Sum of interior angles of a pentagon $= (5 - 2) \times 180° = 540°$ **[B1]**

$114 + (3x + 16) + (4x - 14) + (2x + 34) + 66 = 540$ **[M1]**

$9x + 216 = 540$ leading to $9x = 324$ and $x = 36$ **[A1]**

(c) **(i)** 114 **[B1]**

(ii) Angle $PQR = 3(36) + 16 = 124°$ and angle $WQR = 180° - 124° = 56°$ **[M1]**

56 **[A1]**

(d) Exterior angle $= \frac{180°}{12} = 15°$ **[M1]**

$n = \frac{360}{15}$ **[M1]**

24 **[A1]**

[Back to Question] *[Total Marks: 11]*